10 June 1968
SUPERSEDING
MIL-W-12465B (GL)
8 September 1965

MILITARY SPECIFICATION

WAX, MICROCRYSTALLINE, (ELECTRICAL, INSULATING)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electric insulating microcrystalline wax with fungistatic properties (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

FEDERAL

SS-R-406 - Road and Paving Materials, Methods of Sampling and Testing.
PPP-B-636 - Box Fiberboard.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products: Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 9160

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM Designation):

D176-59 - Testing Solid Filling and Treating Compounds Used for Electrical Insulation.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.- The wax shall be made from such material as to insure compliance with the requirements of this specification. The wax shall not contain mercury in any form and shall be made fungistatic by the addition of a suitable solubilized copper-8-quinolinolate formulation (see 3.10).

3.2 Chemical and physical properties.- The wax shall conform to the requirements of table I when tested as specified in 4.3.6.

TABLE I.- Chemical and physical properties

	Minimum	Maximum
Melting point, ^{o}F.	180	185
Viscosity at 210°F., S.U.S.	70	95
Flash point, Cleveland O.C. ^{o}F.	500	---
Fire point, Cleveland O.C. ^{o}F.	550	---
Acid number, mg. KOH per g.	---	1.5

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TABLE I.- Chemical and physical properties (cont'd)

	Minimum	Maximum
Specific gravity at 60°/60°F.	0.850	---
Loss on heating, percent by weight	---	1.0
Dielectric strength at 86°F., kilovolts per 0.1 inch	80	---

3.3 Stability.- The wax shall show no evidence of sludging or tendency to separate after being heated for 72 hours at 210°F. as specified in 4.3.6.

3.4 Corrosive effect.- Steel, aluminum, and brass shall show no evidence of discoloration or corrosion by the wax, when tested as specified in 4.3.6.

* 3.5 Penetration.- The wax shall have the following needle penetration when tested as specified in 4.3.6.

Total load, g.	Time, sec.	Temperature, °F.	Penetration range, 0.1 mm
100	5	77	2 to 10
50	5	115	25 max.

3.6 Adhesion to metal at low temperatures.- There shall be no evidence of cracks or separation at -5°F. when tested as specified in 4.3.6.

3.7 Water absorption.- The wax shall not absorb more than 0.2 percent of water by weight when tested as specified in 4.3.6.

3.8 Electrolytic corrosion.- The wax shall not corrode bare copper electrodes when tested as specified in 4.3.6.

3.9 Fungus resistance.- There shall be no visible mold growth on the waxed filter paper when tested as specified in 4.3.6.

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3.10 Copper-8-quinolinolate content.- The wax shall be made fungistatic by the addition of a suitable solubilized copper-8-quinolinolate formulation. The addition of the fungistatic agent shall be such that the wax contains a concentration of 0.20 to 0.25 percent of copper-8-quinolinate when tested as specified in 4.3.6. (The required concentration of 0.20 to 0.25 percent refers to the active ingredient, copper-8-quinolinolate, and does not include the solubilizing vehicle.) (See 6.3).

3.11 Workmanship.- The finished wax shall be clean, free from foreign matter and homogeneous in appearance.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.-

4.3.1 Examination of the end item.- The end item shall be examined for the defects in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.5. The lot size shall be expressed in units of 5 to 15 pound cakes of wax as applicable for examination in 4.3.2 and 4.3.3 and in units of shipping containers for examination in 4.3.4.

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4.3.2 Examination of the end item.- The sample unit for this examination shall be one cake.

Examine	Defect
Workmanship	Not homogeneous
	Not clean
	Not free of foreign matter

4.3.3 Examination of the end item for defects in net weight.- The sample unit for this examination shall be one cake of wax as applicable. The lot shall be unacceptable if the average net weight per cake is less than indicated or specified.

* 4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and markings comply with the requirements of Section 5 of this specification. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation shipping containers, fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Not level specified; not in accordance with contract requirements
	Wax not individually wrapped or bagged
	Package not restrained from becoming undone
Packing	Not level specified; not in accordance with contract requirements
	Interior packages not snugly packed within the container
	Inside of shipping container not fitted with taped liner as specified
	Weight of contents of shipping container exceeds 65 pounds
	Container material not as specified; closures not accomplished by specified or required methods or materials
	Bulged or distorted container

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Examine	Defect
Marking	Incorrect; illegible; omitted; incomplete; of improper size, location, sequence or method of application

4.3.5 Inspection levels and acceptable quality levels (AQL's) for examinations.- The inspection levels and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2	I	1.5
4.3.3	S-3	N.A
4.3.4	S-2	2.5

* 4.3.6 Testing of the end item.- The methods of testing specified in Fed. Test Method Std. NO. 791, where applicable, and as listed in table II shall be followed for each lot. For purposes of sampling, the lot shall be expressed in pounds of wax manufactured as a single batch. The sample unit for testing shall be a 3-pound composite obtained by combining equal portions from samples selected at random throughout the lot. The sample size shall be as indicated below. The portions selected for the composite sample shall be melted at a temperature not exceeding 235°F and the mixture subjected to continuous stirring while being cooled to ambient temperature. The composite shall be placed in a clean, dry, sealed container. Care shall be exercised to prevent contamination or alteration of the composite during sampling, compositing, storage and testing. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

Lot size (pounds)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING

END ITEM

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Samp. Unit	Comp. Samp.		Pass or Fail	Numerically to Nearest		
Material	3.1	1/		X	Average of 2		0.5°F.		
Melting point	Table I	1401 2/		X	Average of 2		1.0 sec.		
Viscosity	Table I	304 2/		X	Average of 2		5.0°F.		
Flash point, Cleveland O.C.	Table I	1103 2/		X	Average of 2		5.0°F.		
Fire point, Cleveland O.C.	Table I	1103 2/		X	Average of 2		1 mg.		
Acid number	Table I	4.4.1		X	1		.01		
Specific gravity	Table I	4.4.2		X	Average of 2		0.1 percent		
Loss on heating	Table I	4.4.3	X		Average of 10		1 kilovolt		
Dielectric strength at 86°F., kilovolts	3.3	4.4.5		X	1	X			
Stability	3.4	4.4.6		X	1	X			
Corrosive effect	3.5	4.4.7		X	Average of 2		1.0		
Penetration	3.6	4.4.8		X	3	X			
Adhesion to metal at low temperatures	3.7	4.4.9		X	Average of 2		0.1 percent		
Water absorption	3.8	4.4.10		X	1	X			
Electrolytic corrosion	3.9	4.4.11	X		4	X			
Fungus resistance	3.10	4.4.12		X	Average of 2		.01 percent		
Copper-8-quinolinolate									

1/ Certificate of compliance.- The supplier shall submit to the contracting officer a certificate of compliance that the finished product does not contain mercury in any form, and that the copper fungistatic agent is solubilized copper-8-quinolinolate.

2/ Indicate Fed. Test Method Std. No. 791.

AMXRE Form 598
8 February 1965

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4.4 Test procedure.-

4.4.1 Acid number.- The acid number shall be determined in accordance with Method 5105 of Fed. Test Method Std. No. 791, except that the weighed sample shall be dissolved in 50 milliliters (ml.) of hot xylene before starting the test. If difficulty is encountered in observing the end point of titrations, the electrometric method designated as Method 5106 of Fed. Test Method Std. No. 791 shall be used with the above modification.

4.4.2 Specific gravity.- The specific gravity of the wax shall be determined in accordance with Method 209 of SS-R-406 (displacement method).

4.4.3 Loss on heating.- A representative sample of about 200 g. shall be weighed accurately in a clean porcelain dish permitting an exposed area of the compound of 12 to 15 square inches (about 4 inches in diameter). The dish containing the compound shall be placed in a forced draft oven maintained at 300° ± 5°F. for 4 hours. Calculate and report the percent loss in weight.

4.4.4 Dielectric strength test.- The dielectric strength test shall be determined in accordance with ASTM Designation D176-59.

4.4.4.1 Number of tests.- Unless otherwise specified, 10 tests shall be made at 86° ± 2°F. The average value of breakdown voltage per one-tenth of an inch thickness of material obtained from the specimens tested from each sample unit shall be taken as the dielectric strength of the sample unit.

4.4.5 Stability.- A 200 g. sample of the wax shall be placed in a 400 ml. heat-resistant glass beaker and heated for 72 hours in an oven maintained at 210° ± 5°F. At the end of the heating period, the wax shall be removed and examined visually for evidence of separation of components and sludge.

(Examination may be facilitated by pouring the molten wax into a second beaker and examining the bottom and sides of the first beaker for separated material).

4.4.6 Corrosive effect.- Polished strips of steel, aluminum and brass, each measuring approximately 1/2 by 3 by 1/8 inches, shall be placed separately in three clean test tubes. The test tubes shall be placed in a water bath maintained at 212° ± 2°F. and the melted wax added slowly to each tube until the strips are covered. A vented stopper shall then be used to close each test tube. After remaining in the water bath for 3 hours, the metal strips shall be removed from the test tubes, the wax completely wiped off the strips, and the strips examined visually for discoloration and corrosion. Slight etching shall not be considered as evidence of corrosion.

4.4.7 Penetration.- The penetration of the wax shall be determined in accordance with Method 214 of SS-R-406.

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4.4.8 Adhesion to metal at low temperatures.- A thin coating of the compound, 0.0025 ± 0.0005 inch thick, shall be applied to a clean metal panel of low carbon steel of about 20 gauge. The clean panel shall be heated to 245° ± 5°F. in an oven maintained at that temperature. While at that temperature, the panels shall be immersed in the molten compound maintained at 245° ± 5°F. The panel shall be withdrawn from the molten wax and tilted to one side to allow the excess wax to drip off one corner of the panel. The coated panel shall be set aside to cool to room temperature and then placed in a cold room maintained at minus 5°F ± 0.5°F. for 2 hours. The panel shall be removed and immediately bent through 180 degrees over a half-inch mandrel which had also been maintained at that temperature. This test shall be run in triplicate.

4.4.9 Water absorption.- A sufficient quantity of the compound shall be melted and poured into a mold so that on cooling, a block of material approximately one inch by one inch by two inches is obtained. The sample shall be allowed to solidify and cooled in a desiccator for at least six hours. A block showing any crack after cooling shall not be used for test. The sample shall be accurately weighed and immersed in a liter of distilled water for 24 hours at a temperature of 77° ± 2°F. The sample shall be removed from the water, wiped dry of surface water with a soft rag and weighed accurately. Calculate the percent of water absorbed.

4.4.10 Electrolytic corrosion.- A glass rod approximately 1/4 inch in diameter and 8 inches long shall be flash dipped into the molten material maintained at 15°F. above its complete melting point. Two electrodes, each consisting of a single turn of bare copper wire of 100.5 circular mils area (30AWG), shall be looped around the center of the material and spaced 1/4 ± 1/16 inch apart. Each electrode shall be brought out to opposite ends of the glass rod by winding 3 to 4 turns per inch around the rod. Two such samples shall be placed in a humidity chamber at 70° ± 2°F., and a humidity of above 90 percent for seven days. A potential of 200 to 250 volts D.C. shall be placed on the electrodes while in the humidity chamber. The sample shall be examined at the end of this test and the electrodes shall show no evidence of green corrosion products.

4.4.11 Fungus resistance.- The fungus resistance of the wax shall be determined following the procedure prescribed hereinafter.

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4.4.11.1 Preparation of cultures.-

4.4.11.2 Test organisms.- The following fungi shall be used for testing the fungistatic effectiveness of the wax (see 6.4).

Name of Organisms	American Type Culture Collection No.	Quartermaster Culture Collection No.
Aspergillus niger	9642	QM 386
Aspergillus flavus	9643	QM 380
Penicillium funiculosum	9644	QM 391
Trichoderma sp	9645	QM 365

4.4.11.3 Maintaining stock cultures.- Cultures of the fungi listed in 4.4.11.2 shall be maintained separately on an appropriate medium, such as potato-dextrose agar. The stock cultures shall be kept not more than four months in a refrigerator at a temperature from 3° to 10°C. Subcultures incubated at 28° to 30°C. for 7 to 20 days shall be used in preparing spore suspensions.

4.4.11.4 Spore suspension.- Prepare a spore suspension of each of the four fungi by pouring into one subculture of each fungus a sterile 10-ml portion of water or of a sterile solution containing 0.05 g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125 ml glass-stoppered Erlenmeyer flask containing 45 ml of sterile water. Filter the suspension through a thin layer of sterile glass wool in a glass funnel into a sterile flask in order to remove mycelial fragments. Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid. Resuspend the residue in 50 ml of sterile water and centrifuge. Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with the sterile mineral-salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ± 200,000 spores per ml as determined with a counting chamber. Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension to obtain the final mixed spore suspension. The spore suspension may be prepared fresh each day or may be held in the refrigerator at 3° to 10°C. (37.4° to 50°F.) for not more than 4 days.

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Preparation of mineral-salts solution.- The solution shall contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7 g
Potassium monohydrogen orthophosphate (K_2HOP_4)	0.7 g
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0.7 g
Ammonium nitrate (NH_4NO_3)	1.0 g
Sodium chloride (NaCl)	0.005 g
Ferrous sulfate ($FeSO_4 \cdot 7H_2O$)	0.002 g
Zinc sulfate ($ZnSO_4 \cdot 7H_2O$)	0.002 g
Manganous sulfate ($MnSO_4 \cdot 7H_2O$)	0.001 g
Distilled water	1000 ml

4.4.11.5 Test medium.- The previously sterilized and molten medium shall be poured into sterile 90 mm diameter petri dishes with approximately 25 ml of medium placed in each dish, and allowed to solidify. The test medium shall be the same as specified in 4.4.11.4, except that 15.0 grams agar shall be added.

4.4.11.6 Preparation of test specimens.-

4.4.11.6.1 Conditioning of wax.- Place approximately 300 ml of wax in a low form 600 ml beaker. Heat the container with the wax in a well-ventilated oven maintained at $115^{\circ} \pm 5^{\circ}C$. for 24 hours. Remove the container with the wax from the oven and place it on a hot plate, maintaining the wax at $100^{\circ} \pm 5^{\circ}C$.

4.4.11.6.2 Test specimens.- A sheet of Whatman No. 2 filter paper or approved equal shall be dipped into the molten wax. The impregnated filter paper shall be withdrawn and held under a heating lamp to drain off the excess wax, and cooled to room temperature. The wax coating shall be not greater than 3 mils on each side of the paper. A total of 4 test specimens, each one inch square, shall be cut or punched from the wax coated filter paper for each wax sample tested.

4.4.11.6.3 Inoculation and incubation.- Each of the 4 test specimens prepared as in 4.4.11.6.2 shall be placed on the center of the set agar in separate petri dishes (see 4.4.11.7). The surface of the medium as well as that of the test specimen shall then be inoculated with the composite spore suspension either by spraying the suspension from an atomizer so that the entire surface is moistened by the suspension, or by delivering 0.5 to 1.0 ml of the suspension from a pipette,

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and tilting the dish from side to side in order to moisten the entire surface of the medium and the specimen. With each daily group of tests, each of 3 pieces of filter paper, 1-inch square, shall be placed on set agar in separate petri dishes. These shall be seeded with the spore suspension, the manner indicated above, to serve as controls. The petri dishes shall be incubated at a temperature of 25° to 30°C. at a relative humidity over 90 percent and shall be examined after 14 days incubation.

4.4.11.7 Evaluation of results.-

4.4.11.7.1 Fungus growth in control dishes.- There shall be copious fungus growth on the uncoated filter paper squares in each of 3 control dishes. Absence of such growth requires repetition of the test, and shall not be construed as constituting a failure on the part of the wax.

4.4.11.7.2 Fungus growth in test dishes.- There shall be no visible mold growth on any of the test specimens, as observed with the naked eye. If mold growth is present on only one of the test specimens, the test may be repeated. In order for the wax to be considered fungistatic, all of the test specimens shall pass the test.

4.4.12 Copper-8-quinolinolate determination.- Transfer a 5 g sample to a 400-ml beaker and heat at highest heat of hot plate until the wax is melted. Continue heating at highest heat without a cover glass (to allow free access of air) and add an 11 cm filter paper that has been folded twice and set aflame. The burning part of the filter paper should protrude above the melt. The wax will ignite and burn. After the burning is complete, remove the beaker from the hot plate and allow to cool. Add 30 ml of concentrated nitric acid and 10 ml of perchloric acid (70 percent). Cover with a watch glass and boil down at moderate heat to about 5 ml to destroy the residual organic matter. Allow to cool, and add 300 ml of distilled water and 3 ml of concentrated nitric acid. Electrolyze for copper with stirring for 1 hour at 2 amp. per sq. dm., using a platinum anode and a tared platinum gauze cathode. Without interrupting the current, lower the beaker while washing the cathode with a stream of water from a wash bottle. Dip the cathode in distilled water and then in 95 percent ethyl alcohol. Dry the cathode in an oven at 105°C. for 5 minutes, cool, weigh the deposit as metallic copper.

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Calculate as follows:

$$\text{Copper-8-quinolinolate, percent} = \frac{553.5 \times \text{grams of copper}}{\text{grams of sample}}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.- Wax, put up in cake form of 5 to 15 pounds, shall be individually wrapped or bagged in glassine paper or other grease resistant material that is compatible with the product. The package shall be restrained from becoming undone.

5.1.2 Level C.- Wax shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A.- Wax, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a taped liner conforming to type CF, class weather-resistant variety DW, grade V15c of PPP-F-320. Each shipping container shall be closed, waterproofed and reinforced in accordance with the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

* 5.2.2 Level B.- Wax, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a taped liner conforming to class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification. The weight of the contents of each shipping container shall not exceed 65 pounds.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification. The shipping container material may also be grade V4s of PPP-F-320.

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* 5.2.3 Level C.- Wax, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The wax covered by this specification is intended for the treatment of electrical equipment to protect against fungus attack and visible fungus growth. It is not suitable for coating printed circuit board assemblies. It is not intended for use where temperatures exceed 165°F.

* 6.2 Ordering data.- Procurement documents should specify the following:

a. Title, date and number of this specification

b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

c. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Copper-8-quinolinolate.- The specified test is for the purpose of ascertaining that the fungicide is properly blended with the wax and is not to be considered a performance test for evaluating fungus resistant agents.

6.4 Test organisms.- Cultures of the fungi listed under 4.4.11.2 may be obtained from the American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852, or for service use from the Pioneering Research Laboratory, U. S. Army Natick Laboratories, Natick, Massachusetts 01762.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

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